

Progress test 2

1. Multiple integral and line integral

- (a) Find the centre of mass of the rectangle given by $[0, 1] \times [0, 1]$ if the mass density is e^{x+y} . [20%]

- (b) Consider a graph given by $y = f(x)$ for $x \in [a, b]$. Show that the length of the graph is given by [20%]

$$\int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

- (c) Let an oriented closed curve c be composed of two curves, given by $c_1(t) = (t, 2t^3)$ for $t \in [0, 1]$ and $c_2(s) = (s, 2s)$ for $s \in [0, 1]$, respectively. Find $\int_c \mathbf{F} \cdot d\mathbf{s}$ if a vector field is given by $\mathbf{F}(x, y) = (x^2, y^2)$. [20%]

- (d) Let a curve p be given by $p(t) = (1 - e^{t-1}, \pi \sin^4(\pi(t-1)/2))$ for $t \in [1, 2]$. Find $\int_p \mathbf{G} \cdot d\mathbf{s}$ if $\mathbf{G} = 2x \cos y \mathbf{i} - x^2 \sin y \mathbf{j}$. [40%]

$$\begin{aligned}
 \text{(a) centre of mass} &= \frac{\int_A \rho dA}{\int_A dA} \\
 &= \frac{\int_0^1 \int_0^1 e^{x+y} dx dy}{\int_0^1 \int_0^1 dx dy} \\
 &= \frac{\int_0^1 e^{x+1} + 1 dy}{\int_0^1 dy} \\
 &= \frac{e^2 + 1 - e}{1} \\
 &= e^2 - e + 1
 \end{aligned}$$

$$\begin{aligned}
 \text{a) } x_{CG} \cdot M &= \int_0^1 \int_0^1 x e^{x+y} dx dy \\
 &= \int_0^1 [x e^{x+y} - e^{x+y}]_0^1 dy \\
 &= \int_0^1 e^y dy = e - 1
 \end{aligned}$$

$$\begin{aligned}
 y_{CG} \cdot M &= \int_0^1 \int_0^1 y e^{x+y} dx dy \\
 &= e - 1
 \end{aligned}$$

$$\begin{aligned}
 M &= \int_0^1 \int_0^1 e^{x+y} dx dy \\
 &= \int_0^1 [e^{x+y}]_0^1 dy = \int_0^1 e^{y+1} - e^y dy \\
 &= [e^{y+1} - e^y]_0^1 = e^2 - e - e + 1 = (e-1)^2
 \end{aligned}$$

$$\therefore CG = \left(\frac{1}{e-1}, \frac{1}{e-1} \right)$$

$$(b) \text{ length} = \int_c \left\| \frac{dc}{dt} \right\| dt$$

$$\text{for } c = (x, f(x))$$

$$\text{length} = \int_a^b \left\| (1, f'(x)) \right\| dt$$

$$= \int_a^b \sqrt{1 + (f'(x))^2} dt$$

$$(c) \int_c F \cdot ds = \int_{c_1} F \cdot ds - \int_{c_2} F \cdot ds$$

$$= \int_0^1 (t, 4t^6) \cdot (1, 6t^5) dt$$

$$- \int_0^1 (s^2, 4s^3) \cdot (1, 2) ds$$

$$= \int_0^1 t + 24t^6 dt - \int_0^1 4s^2 ds$$

$$= \left[\frac{1}{2}t^2 + \frac{8}{3}t^7 \right]_0^1 - \left[\frac{4}{3}s^3 \right]_0^1$$

$$= 3 - 3 = 0$$

$$\begin{aligned} \text{(d)} \quad \nabla \times \mathbf{G} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x \cos y & -x^2 \sin y & 0 \end{vmatrix} \\ &= (0, 0, (-2x \sin y + 2x \sin y)) = \mathbf{0} \end{aligned}$$

$$\therefore \nabla g = \mathbf{G}$$

$$\therefore g_x = x^2 \cos y + h(y, z) + C_x$$

$$g_y = x^2 \sin y + h(x, z) + C_y$$

$$g_z = h(x, y) + C_z$$

$$\therefore g = x^2 \cos y + C$$

$$\therefore \int_C \mathbf{G} \cdot d\mathbf{s} = \int_C \nabla g \cdot d\mathbf{s}$$

$$= g(c(b)) - g(c(a))$$

$$= g(1-e, \pi) - g(0, 0)$$

$$= (1-e)^2 \cos \pi + C - (0)^2 - C$$

$$= -(1-e)^2$$